# Justin Liu

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## Education

**Vanderbilt University** **Aug 2026 - May 2029**
*B.S. in Computer Science* *Nashville, TN*
**Brooklyn College** **Aug 2025 - May 2025**
*Transfer GPA: 3.8* *Brooklyn, NY*

## Technical Skills

**Languages**: Bash, Java, JavaScript, Python, SQL
**Frameworks & Libraries**: Express.js, NumPy, Pygame, React.js
**Developer Tools**: Blender, Docker, Git, Figma, Node.js, Stripe

## Experience

**Undergraduate Research Assistant** **Aug 2026 – Present**
*Vanderbilt University* *Nashville, TN*

- Operated Bambu and nScrypt 3D printers while collecting synchronized IMU, acoustic, and camera sensor data
- Calibrated a fisheye camera from grid images to estimate intrinsic and distortion parameters and generate pixel-level undistortion mappings
- Developed pixel remapping and perspective correction to geometrically align camera footage for 3D-print monitoring

**Software Engineering Intern** **Mar 2025 – July 2025**
*TruthBridge AI* *Remote*

- Built an interactive search filter using React.js, enabling users to run custom queries and retrieve data across 5,000+ colleges from an Airtable database
- Developed nine search controls, live result counts, and college news displays

**Software Engineering Intern** **Apr 2024 – May 2024**
*NYCDOE Division of Instructional and Information Technology* *NYC, NY*

- Collaborated with the NYC Department of Education's DIIT branch to achieve a successful website redesign, enhancing user experience and accessibility
- Presented redesign recommendations to senior leadership, resulting in adoption of 20+ proposed features

## Projects

**GymCoder** | *Docker, Express.js, Node.js, Stripe, Google OAuth, Pyodide* **Apr 2026 – Present**

- Built a full-stack coding platform supporting Python, JavaScript, and Java execution, custom test cases, and progress tracking
- Implemented authentication, subscription billing, and persistent user state using Node.js, Express.js, SQLite, Google OAuth, and Stripe

**Billiards Shot Solver** | *Blender, NumPy, Pygame* **Sep 2026 – Present**

- Built a physics-based billiards simulator modeling ball collisions and cushion rebounds across configurable table layouts
- Developed a numerical search algorithm to compute angle, power, and spin for user-selected shots
- Created dimensionally accurate 2D and 3D pool table visualizations using Blender and real-world measurements